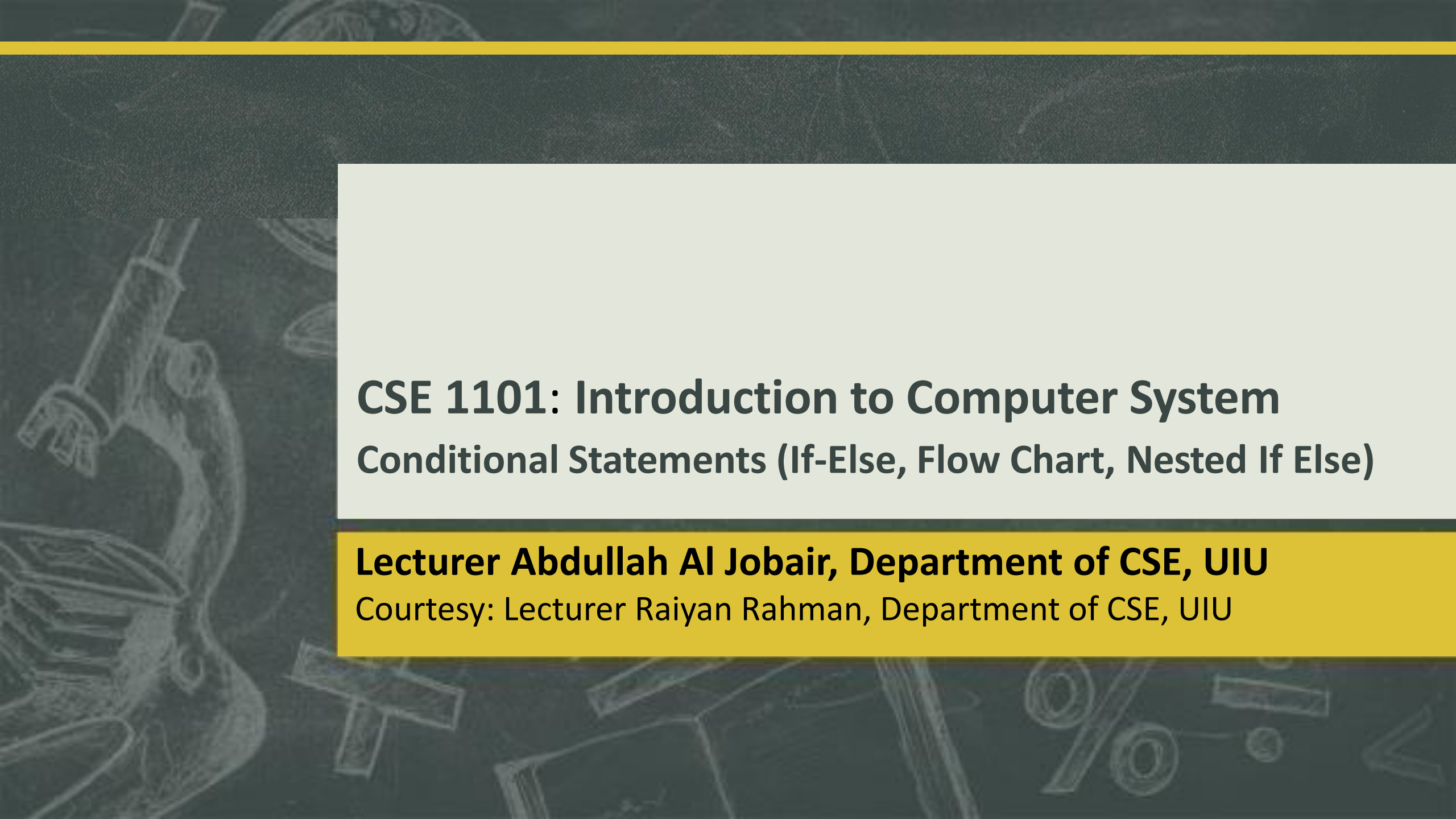




CSE 1101: Introduction to Computer System

Conditional Statements (If-Else, Flow Chart, Nested If Else)

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Courtesy: Lecturer Raiyan Rahman, Department of CSE, UIU

Control Statement

- **Control statements** enable us to specify the flow of program control; ie, the order in which the instructions in a program must be executed.
- There are four types of control statements in C:
 - Decision making statements (example – if-else statement)
 - Selection statements (example – switch-case statement)
 - Iteration statements (example – loop statement)
 - Jump statements (example – goto, break, continue statement)

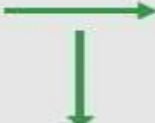
TYPES OF CONTROL STATEMENTS

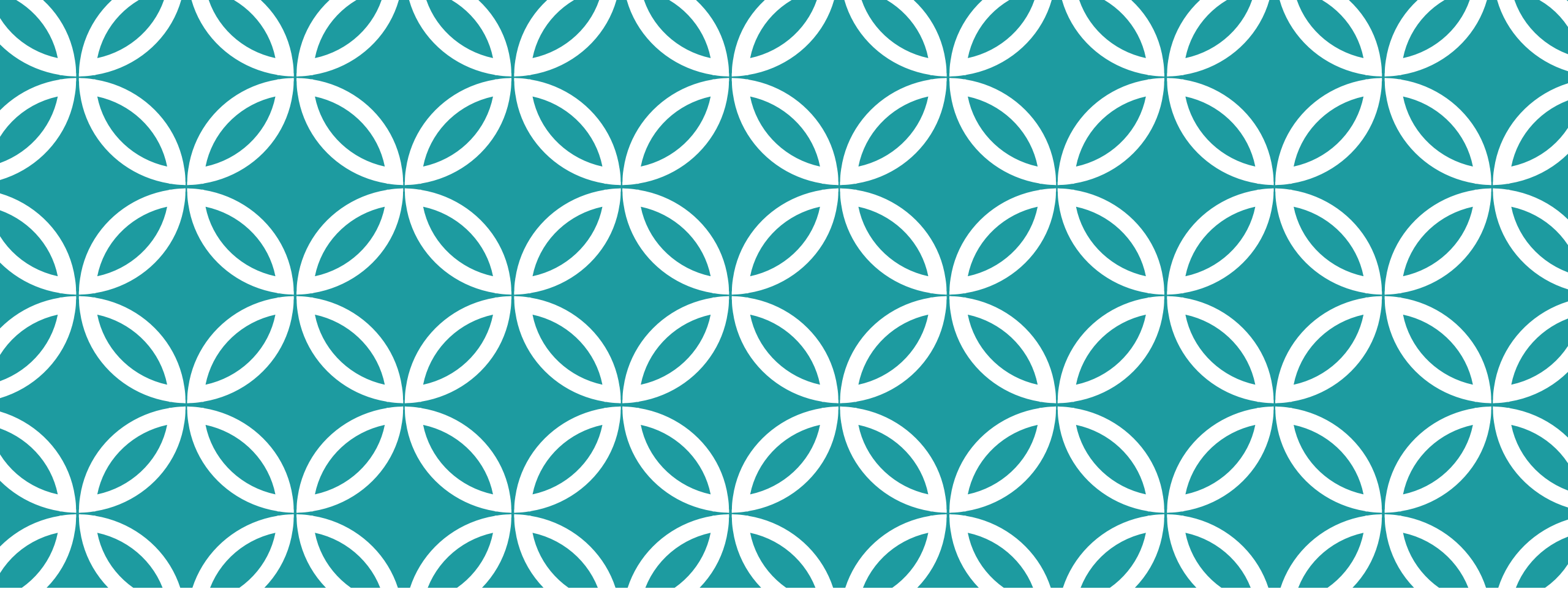
Type	When to Use	Example Case
A single If	Just need to check one condition	Bill checking software. You only except positive bill. Check if the input is >0.
Sequence of only "If"s	Need to check multiple condition one after another (usually on same vars, might be true for multiple cases)	Check seq_of_ifs.c
If and else	Only two conditions, only one of them might be true. Second condition doesn't even need to be checked.	Simple pass fail calculator. If number>=33 passed. Otherwise Failed.
If- else-if- else Ladder	Only one condition out of multiple may be true. Check them one after another. The last or default condition is not checked and written with an "else"	Check if_else_grade.c

If-else Statement

- The if statement may be implemented in different forms.
 1. Simple if statement
 2. Sequence of if statement
 3. if.....else statement
 4. elseif ladder
 5. Nested if.....else statement

Flow Chart Symbols

Name	Symbol	Usage
Start or Stop		The beginning and end points in the sequence.
Process		An instruction or a command.
Decision		A decision, either yes or no.
Input or Output		An input is data received by a computer. An output is a signal or data sent from a computer.
Connector		A jump from one point in the sequence to another.
Direction of flow		Connects the symbols. The arrow shows the direction of flow of instructions.



ONLY “IF”



Simple If Statement

- **Syntax:**

```
if(condition){  
    statement 1;  
    statement 2;  
}
```

- The if statement evaluates the condition inside the parenthesis ().
- If the condition is **true**, statements **inside the body of if are executed**.
- If the condition is **false**, statements **inside the body of if are not executed**.

Syntax - Only “If”

-Practical example:

if it is not a holiday, we will go to UIU.

-Format:

```
if(check if it's holiday)  
    printf("Let's not go to UIU!");
```

-Format:

```
if(expression)  
    statement;
```



No Semicolon

Syntax - Multi Statement If

- Format:

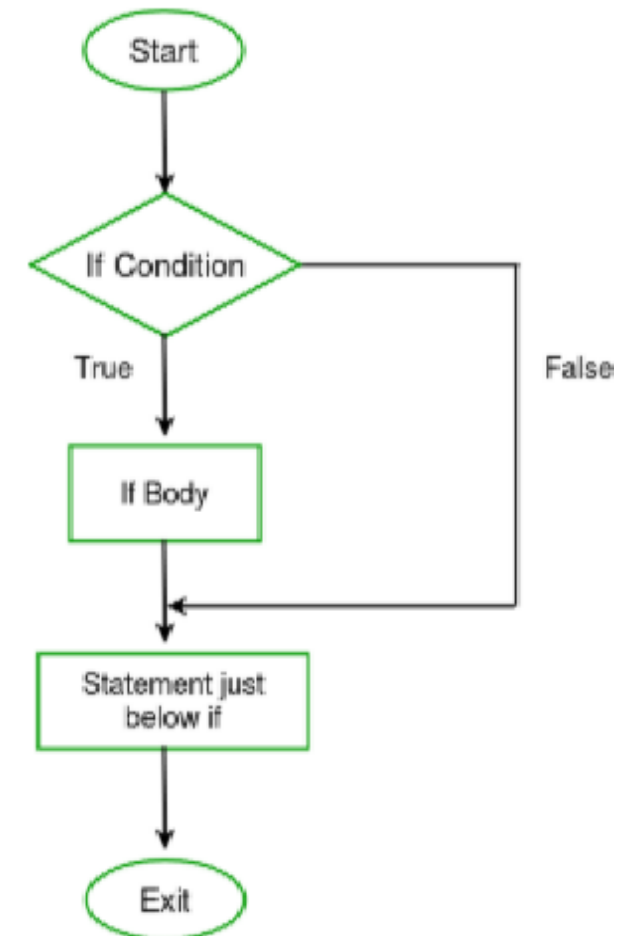
```
if(check if it's raining)
{
    statement1;
    statement2;
    ... .. . . . ;
    statement n;
}
```

Simple If Statement Example

```
#include<stdio.h>
int main() {
    int i;
    printf("Enter an integer grater than 10\n");
    scanf("%d",&i);
    if(i>10){
        printf("You followed the instruction correctly\n");
    }
    printf("program ends");
}
```

Simple If Statement Example using Flow Chart

```
#include<stdio.h>
int main() {
    int i;
    printf("Enter an integer grater than 10\n");
    scanf("%d",&i);
    if(i>10){
        printf("You followed the instruction correctly\n");
    }
    printf("program ends");
}
```



Syntax - Only “If” - An example

```
#include<stdio.h>
void main()
{
    int var;
    scanf(" %d", &var);
    if(var > 0)
        printf("%d is a Positive Number", var);
}
```

Syntax - Multi line “If” - An example

```
#include<stdio.h>
```

```
void main()
```

```
{
```

```
    int var;
```

```
    scanf(“ %d”, &var);
```

```
    if(var > 0)
```

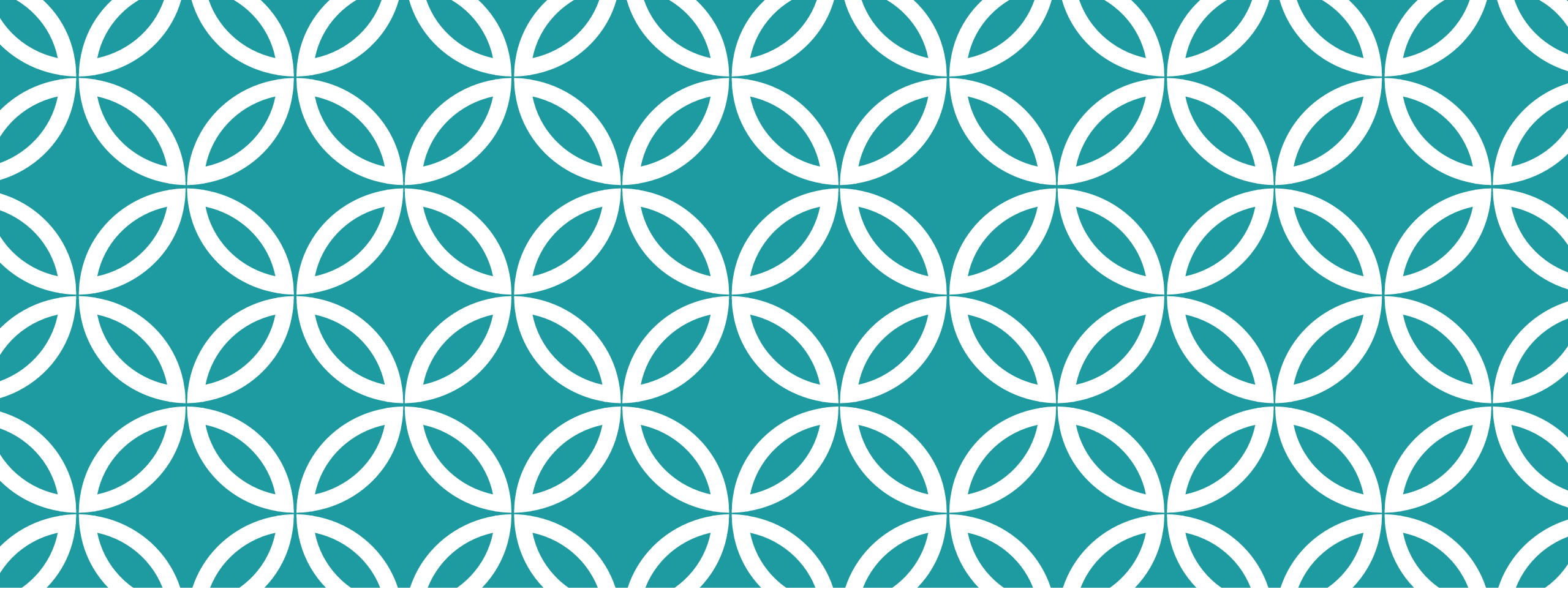
```
    {
```

```
        printf(“The number is a Positive Number”);
```

```
        printf(“\ nThe number is: %d”, var);
```

```
    }
```

```
}
```



SEQ OF “IF” S

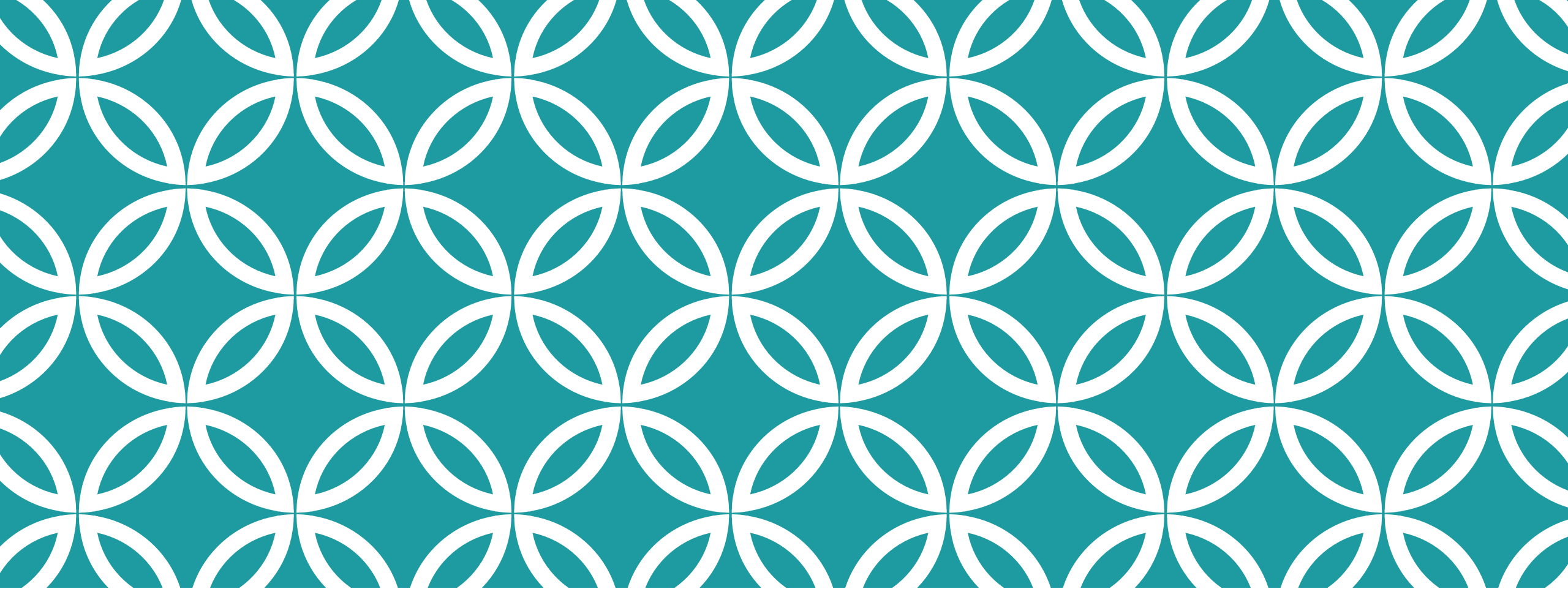
Syntax - Seq of “If”s - An example

```
#include<stdio.h>
void main()
{
    int var;
    scanf(" %d", &var);
    if(var > 0)
        printf("%d is a Positive Number", var);

    if(var < 0)
        printf("%d is a Negative Number", var);
}
```

Syntax - Seq of If Ladder - An example

[Check out seq_of_ifs.c](#)



IF ELSE



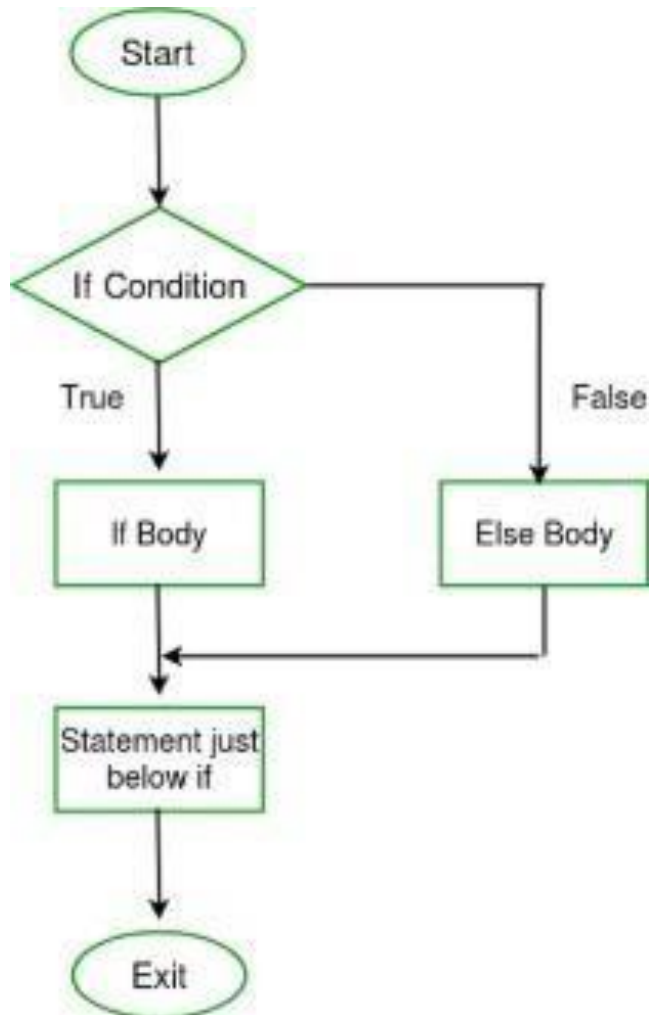
If-else Statement

- **Syntax:**

```
if(condition){  
    statement 1;  
    statement 2;  
}  
else{  
    statement 1;  
    statement 2;  
}
```

- The if statement evaluates the condition inside the parenthesis ().
- If the condition is **true**, statements **inside** the body of **if** **are executed** and statement **inside** the body of **else** is **skipped**.
- If the condition is **false**, statements **inside** the body of **else** **are executed** and statement **inside** the body of **if** is **skipped**.

If-else Statement using Flow Chart



- The if statement evaluates the condition inside the parenthesis ().
- If the condition is **true**, statements **inside** the body of **if** **are executed** and statement **inside** the body of **else** is **skipped**.
- If the condition is **false**, statements **inside** the body of **else** **are executed** and statement **inside** the body of **if** is **skipped**.

If-else Statement Example 1

■ **Question:** Take input of a number and **print** its **absolute value**. Solve the problem using Flow Chart first.

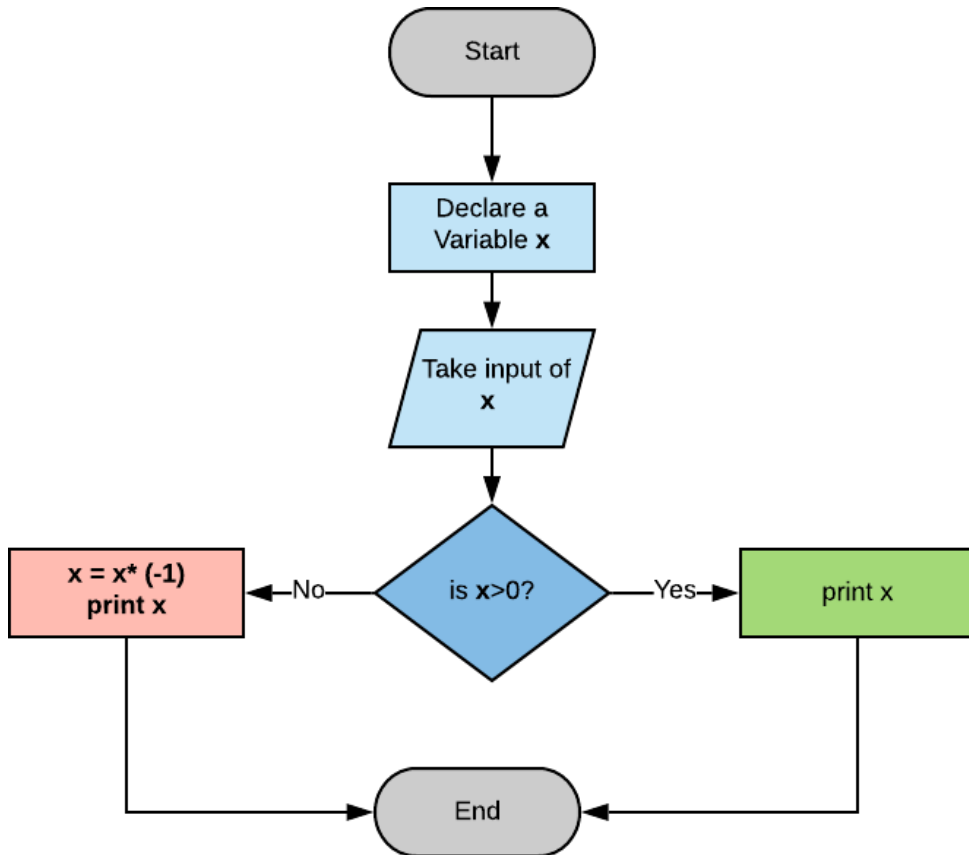
■ **Steps:**

1. First we have to declare a variable **x**
2. Take input of that variable **x**
3. Check if it is greater than zero
4. If yes, then do nothing, just print the value
5. If no, then multiply it by -1 and print the value

If-else Statement Example 1 cntd.

■ Steps:

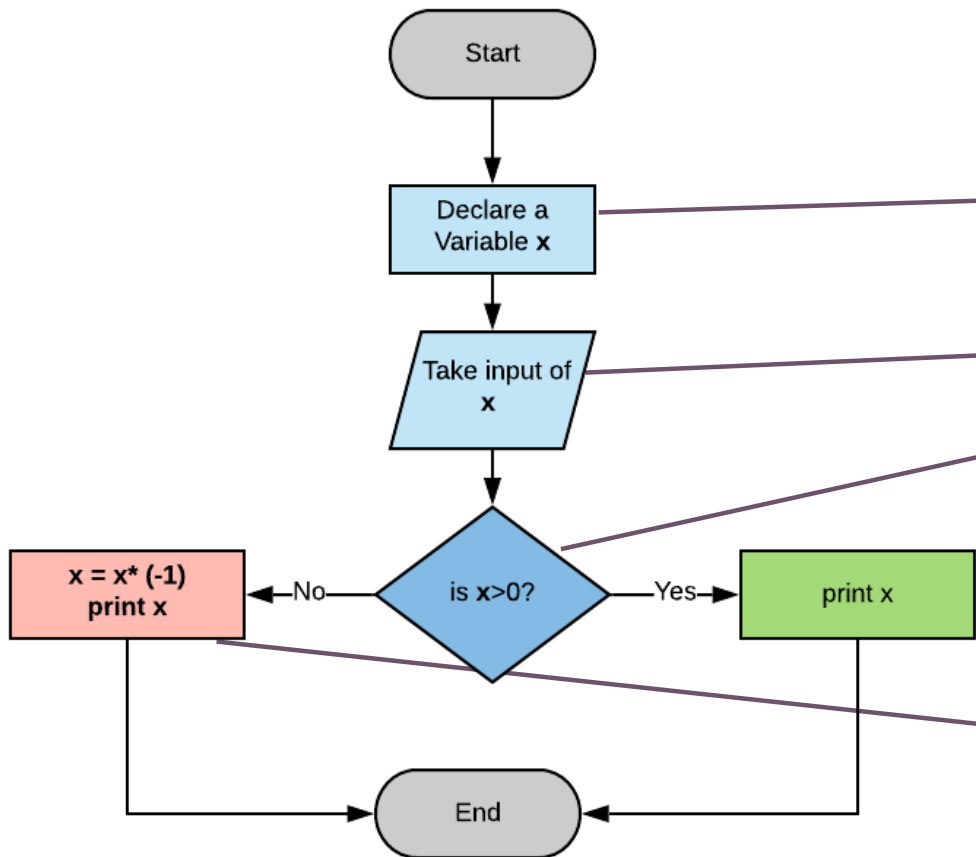
1. First we have to declare a variable **x**
2. Take input of that variable **x**
3. Check if it is grater than zero
4. If yes, then do nothing, just print the value
5. If no, then multiply it by -1 and print the value



If-else Statement Example 1 cntd.

■ Code Representation:

```
#include<stdio.h>
int main() {
    int x;
    printf("Enter an integer \n");
    scanf("%d",&x);
    if(x>0){
        printf("Absolute value is: %d\n",x);
    }
    else{
        x= x*(-1);
        printf("Absolute value is: %d",x);
    }
}
```



If-else Statement Example 2

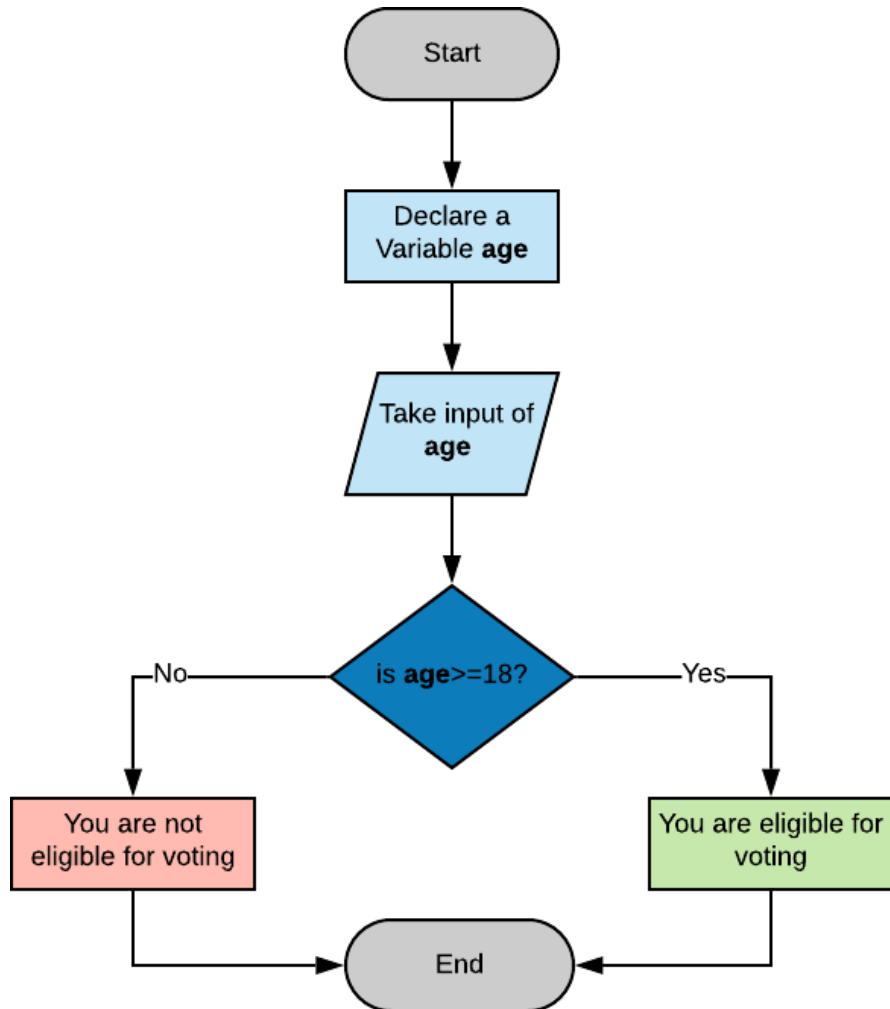
- **Question:** Take input of your age and **test** whether you are **eligible for voting** or **not eligible**. Solve the problem using Flow Chart first.

- **Note:** to be eligible for voting your age has to be atleast 18

- **Steps:**

1. First we have to declare a variable **age**
2. Take input of **age**
3. Check if **age** is grater than equal 18
4. If yes, then you **are eligible**
5. If no, then you **are not eligible**

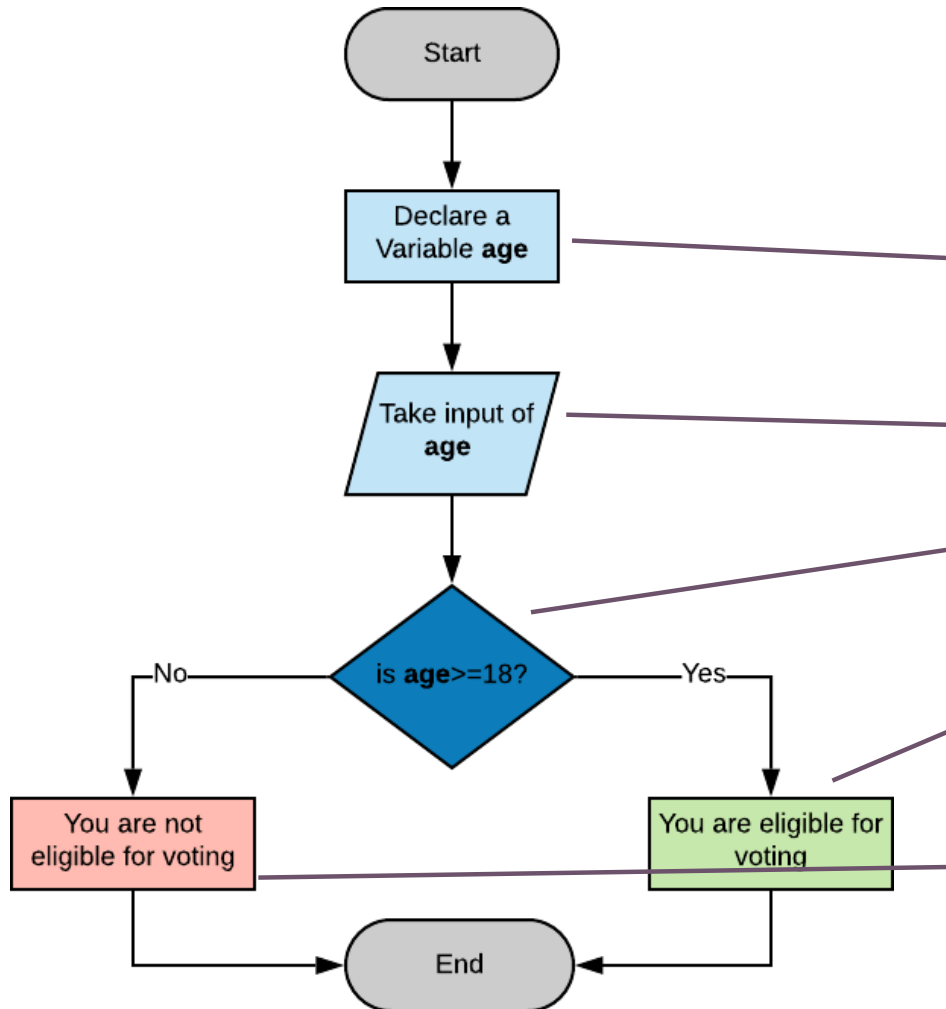
If-else Statement Example 2 cntd.



■ Steps:

1. First we have to declare a variable **age**
2. Take input of **age**
3. Check if **age** is grater than equal 18
4. If yes, then you **are eligible**
5. If no, then you **are not eligible**

If-else Statement Example 2 cntd.



■ Code Representation:

```
#include<stdio.h>
int main() {
    int age;
    printf("Enter your age \n");
    scanf("%d",&age);
    if(age>=18){
        printf("You are eligible for voting\n");
    }
    else{
        printf("You are not eligible for voting");
    }
}
```

Syntax - If-Else

-Practical example:

if x is divisible by 2, then x is an even number.

otherwise x an odd number

-Format:

if(*expression*) *statement1*;

else *statement2*;

-Format:

if(*expression*)

statement1;

else

statement2;

← No Semicolon

← No Semicolon

SAMPLE PROGRAM- IF ELSE

```
#include<stdio.h>
void main()
{
    int var;
    scanf(" %d", &var);
    if(var > -1)
        printf("%d is a Positive Number", var);
    else
        printf("%d is a Negative Number", var);
}
```

➤ Format:

if(*expression*)

{

statement1;

statement2;

.....

}

else

statement 3;

IF ELSE (Con't)

➤ Format:

```
if(expression)  
    statement1;  
  
else  
{  
    statement2;  
    .....  
    statement n;  
}
```

IF ELSE (Con't)

➤ Format:

```
if(expression)
{
    statement1;
    statement2;
    .....
}
else
{
    statement4;
    .....
    statement n;
}
```

IF ELSE (Con't)

➤ Format:

```
if(expression)
{
    statement 1;
}
```

```
statement m;
```

```
else
```

```
{
    statement4;
    .....
    statement n;
}
```

//Wrong

//mismatched if else

Exercise 1 – Multi Line if-else

Problem:

Take buying price and selling price of a product as input and find out the percentage of the profit or loss of the product.

Sample Input 1:

Buy: 10

Sell: 15

Sample Output 1:

Profit 50%

Sample Input 2:

Buy: 40

Sell: 20

Sample Output 2:

Loss 50%

Solution: [check out profit_loss.c](#)



```
#include<stdio.h>
```

```
int main(){  
    double buying_price, selling_price,  
        profit, loss;  
  
    printf("insert buying price \n");  
    scanf("%lf", &buying_price);  
    printf("insert selling price \n");  
    scanf("%lf", &selling_price);
```

```
    if(buying_price < selling_price){  
        profit= selling_price - buying_price;  
        printf("Profit %lf", profit/buying_price*100);  
    }  
    else{  
        loss = buying_price - selling_price;  
        printf("Loss %lf", loss/buying_price*100);  
    }  
}
```

Logical Operator combined

Symbol	Operator_Name
&&	AND
	OR
!	NOT

Exercise 2

Problem:

Take a character as input and determine whether it is a vowel or not.

Sample Input 1:

A

Sample Output 1:

Vowel

Sample Input 2:

B

Sample Output 2:

Consonant

Solution: [check out vowel.c](#)

Exercise 3

Problem:

Take a year as input and find out whether it is leap year (check the leap year [algorithm](#)) or not.

Sample Input 1:

2012

Sample Output 1:

Leap Year

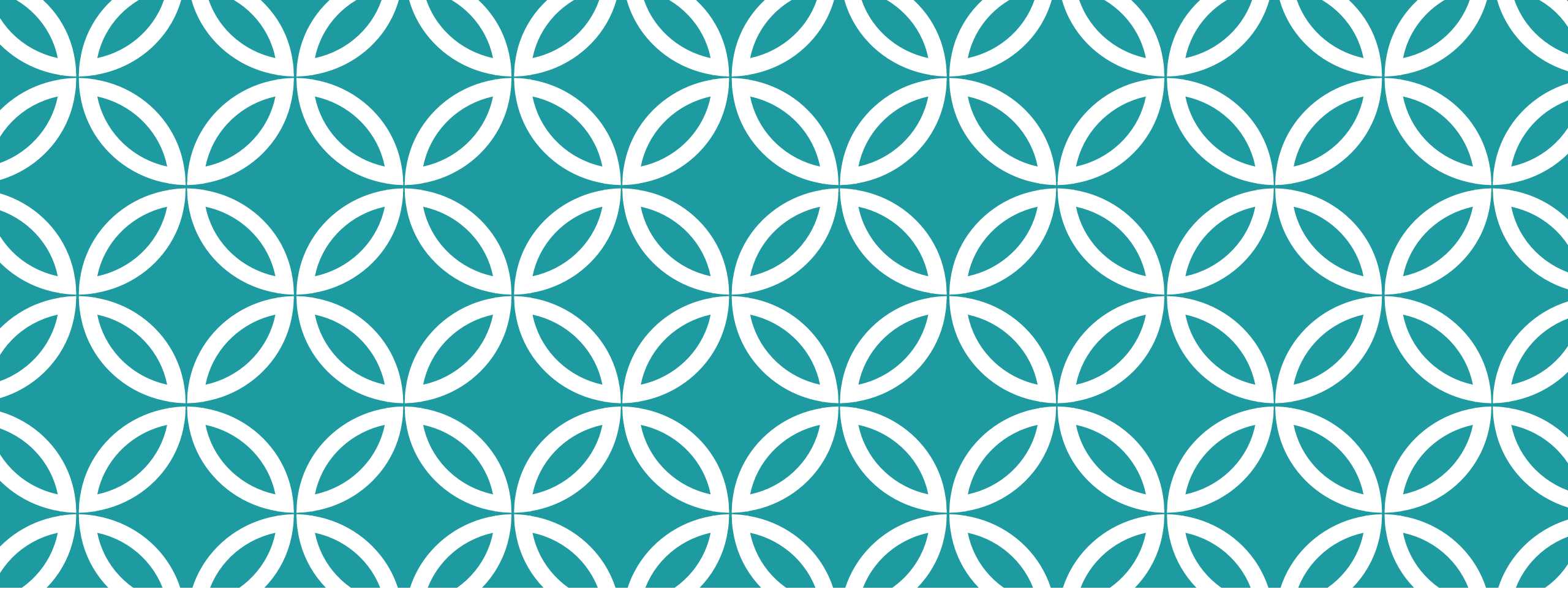
Sample Input 2:

2010

Sample Output 2:

Not Leap Year

Solution: [check out if_else_leap_year.c](#)



“IF-ELSE IF - ELSE” LADDER



elseif ladder statement

```
if (condition1) {  
    // If condition 1 is true then  
    // execute this and skip other if  
}  
else if (condition2) {  
    // If condition 1 is false and  
    // condition 2 is true then  
    // execute this and skip other if  
}  
else {  
    // If no expressions are true then  
    // execute this skipping all other.  
}
```

- The else..if statement is useful when you need to check multiple conditions within the program.

elseif ladder statement



Format:

if(expression)

statement1;

else if(expression)

statement2;

.....

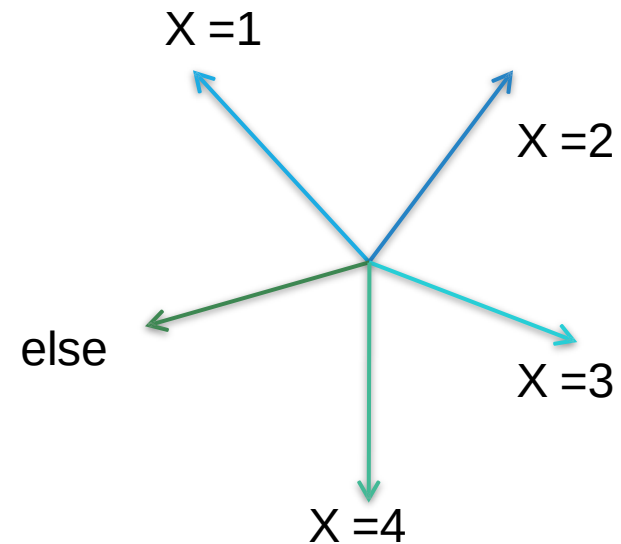
... ..

else if(expression)

statement_n;

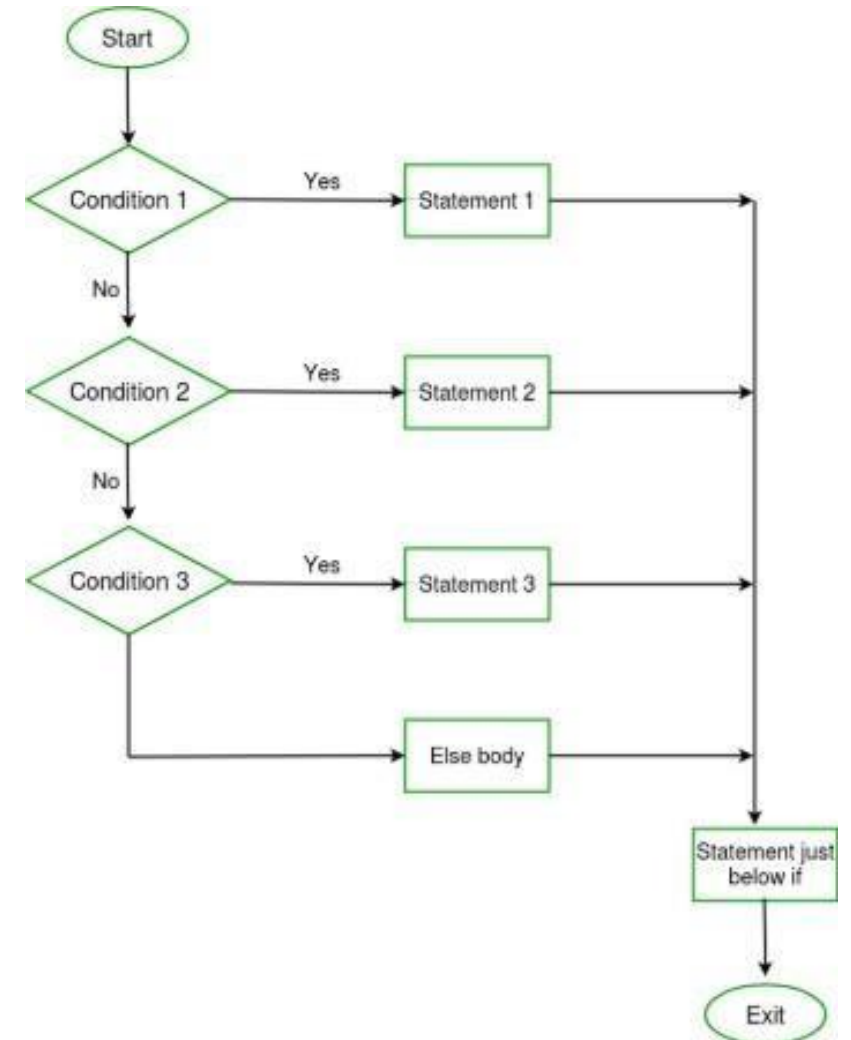
else

statement_x;



elseif ladder statement using Flow Chart

```
if (condition1) {  
    // If condition 1 is true then  
    // execute this and skip other if  
}  
else if (condition2) {  
    // If condition 1 is false and  
    // condition 2 is true then  
    // execute this and skip other if  
}  
else if(condition3){  
    // If condition 1 and 2 is false and  
    // condition 3 is true then  
    // execute this and skip other if  
}  
else  
    // If no expressions are true then  
    // execute this skipping all other.  
}
```



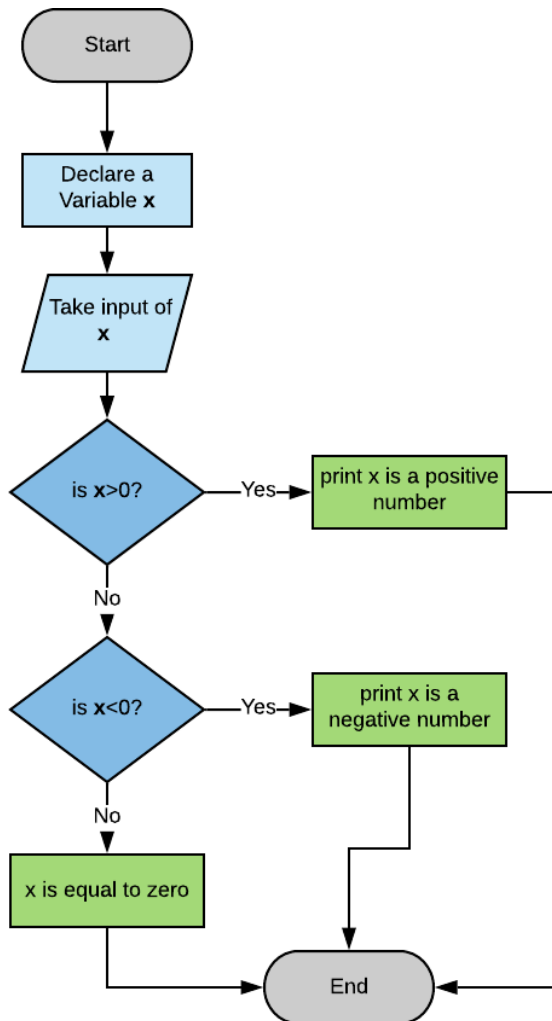
elseif ladder statement Example 1

■ **Question:** Take input of a number and **test** whether it is a **positive number** or **negative number**. Solve the problem using Flow Chart first.

■ **Steps:**

1. First we have to declare a variable
2. Take input of that variable
3. Check if it is greater than zero
4. If yes, then positive number
5. If no, then check if it is less than zero
 1. If yes, then negative number
 2. If no, then the number is zero

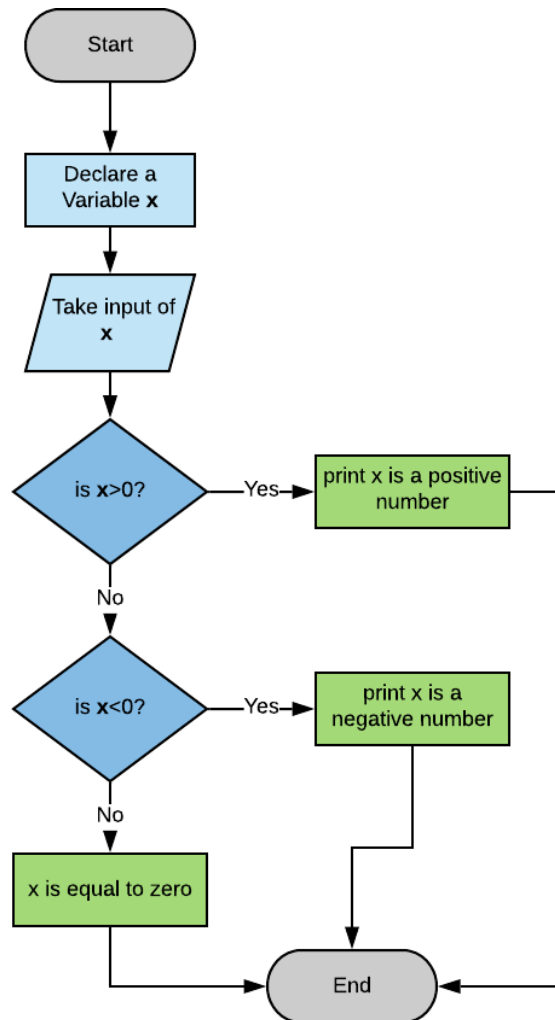
elseif ladder statement Example 1



■ Steps:

1. First we have to declare a variable
2. Take input of that variable
3. Check if it is grater than zero
4. If yes, then positive number
5. If no, then check if it is less than zero
 1. If yes, then negative number
 2. If no, then the number is zero

elseif ladder statement Example 1



▪ Code Representation:

```
#include<stdio.h>
int main() {
    int x;
    printf("Enter an integer \n");
    scanf("%d",&x);
    if(x>0){
        printf("x is a positive number\n");
    }
    else if(x<0){
        printf("x is a negative number");
    }
    else{
        printf("x is equal to zero");
    }
}
```

elseif ladder statement Example 2

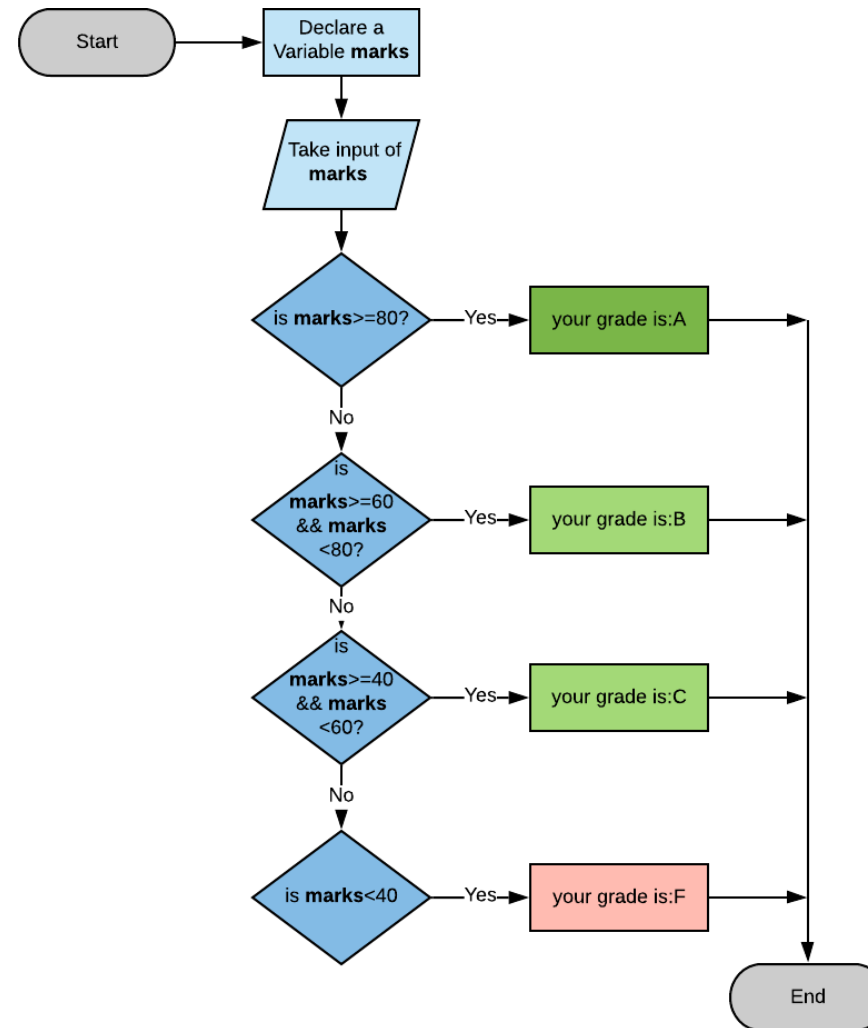
▪ **Question:** Take **input** of your **marks** in CSE 1101 and **print** your **grade**. Solve the problem using Flow Chart first. grading system is given below:

Marks	Grade
grater than or equal 80	A
60 to 79	B
40 to 59	C
Below 40	F

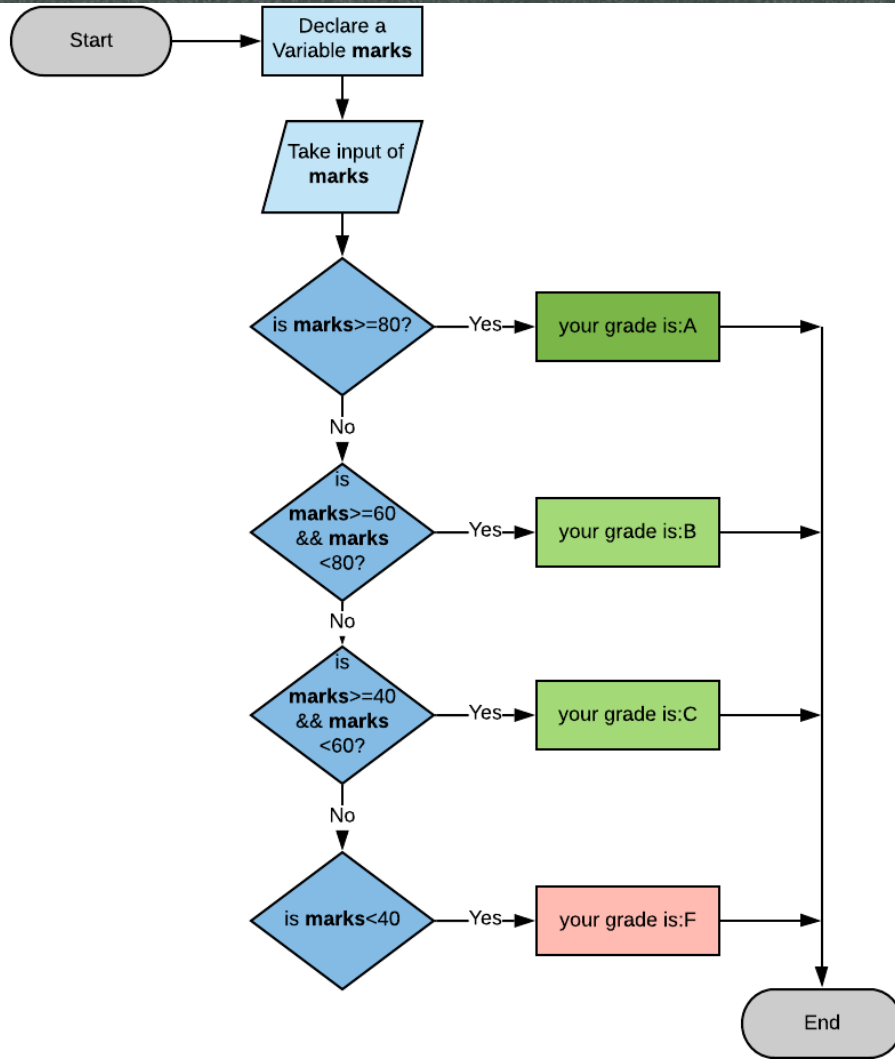
▪ Steps:

1. First we have to declare a variable **marks**
2. Take input of **marks**
3. Check if **marks** in various ranges

elseif ladder statement Example 2



elseif ladder statement Example 2



Code Representation:

```
#include<stdio.h>
int main() {
    int marks;
    printf("Enter your marks \n");
    scanf("%d",&marks);
    if(marks>=80){
        printf("Your grade is:A\n");
    }
    else if(marks>=60 && marks<80){
        printf("Your grade is:B\n");
    }
    else if(marks>=40 && marks<60){
        printf("Your grade is:C\n");
    }
    else if(marks<40){
        printf("Your grade is:F\n");
    }
}
```

EXERCISE 4:

Take one student's number as input and find out the grade of the mentioned student.

Number	Grade
80-100	A+
75-79	A
70-74	A-
65-69	B+
60-64	B
55-59	B-
50-54	C+
45-49	C
40-44	D
<40	F

Sample Input 1:

Number: 76

Sample Output 1:

Grade: A

Sample Input 2:

Number: 44

Sample Output 1:

Grade: D

Solution: [check out if-else_grade.c](#)

```
#include<stdio.h>
```

```
void main()
```

```
{    int a;
```

```
    scanf(" %d", &a);
```

```
    if(a >= 80)
```

```
        printf("A+");
```

```
    else if(a >= 75 && a<80)
```

```
        printf("A");
```

```
    else if(a >= 70 && a<75)
```

```
        printf("A-");
```

```
    else if(a >= 65 && a<70)
```

```
        printf("B+");
```

```
}
```

```
    else if(a >= 60 && a<65)
```

```
        printf("B");
```

```
    else if(a >= 55 && a<60)
```

```
        printf("B-");
```

```
    ... ..
```

```
    ...
```

```
    ... ..
```

```
    ...
```

```
    else if(a >= 40 && a<45)
```

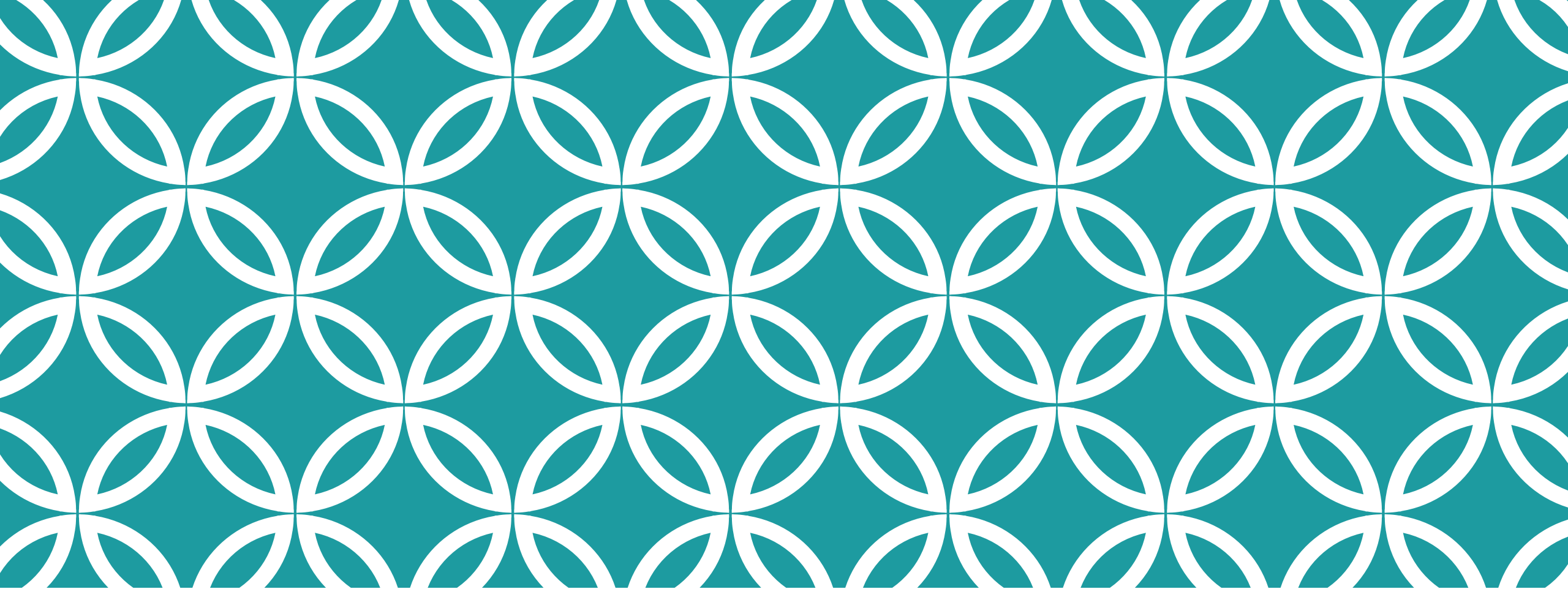
```
        printf("D");
```

```
    else
```

```
        printf("F");
```

Task(Draw flow chart & then write code)

1. Take input of a number and test whether it is an **odd number** or **even number**
2. Take input of a number and print the absolute value of it
3. Take input of 2 numbers and find the **maximum number**
4. Take input of two integer numbers a, b and show whether **a** is **divisible by b** or not.
5. Take input of a character and print whether it is **uppercase** or **lowercase**
6. Take input of two numbers **a, b**; and print if a is equal to b or a grater than b or a less than b



NESTED IF - ELSE

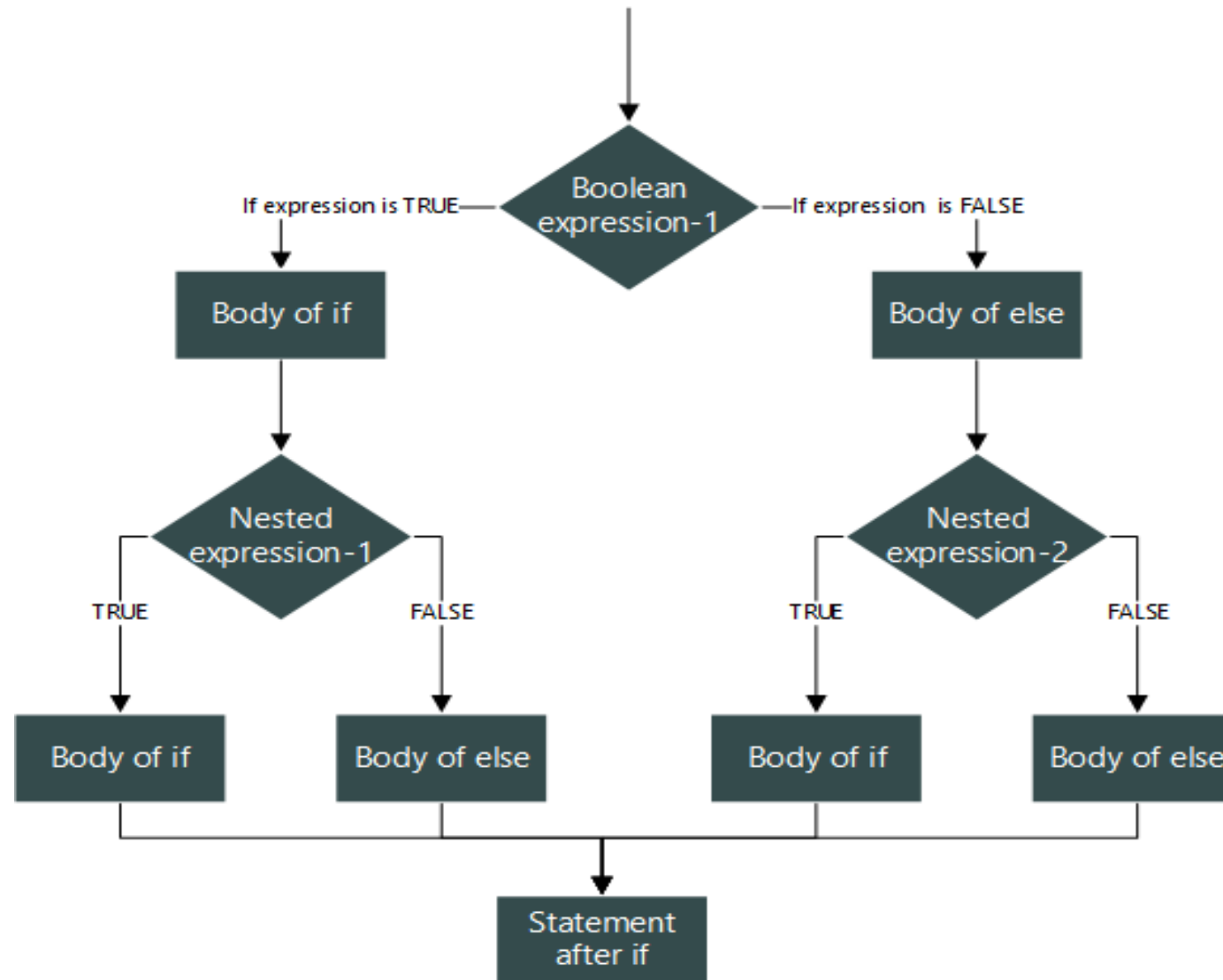


NESTED IF-ELSE

- When an if else statement is present inside the body of another “if” or “else” then this is called nested if else.
- We can use this technique when we need to check further conditions after checking a condition.

```
if(condition) {  
    //Nested if else inside the body of "if"  
    if(condition2) {  
        //Statements inside the body of nested "if"  
    }  
    else {  
        //Statements inside the body of nested "else"  
    }  
}  
else {  
    //Statements inside the body of "else"  
}
```

NESTED IF-ELSE STATEMENT FLOW CHART



NESTED IF-ELSE EXECUTION SEQUENCE

- This condition will be evaluated first. →

- If the condition is false, statements inside the body of else are executed

```
if(condition) {  
    //Nested if else inside the body of "if"  
    if(condition2) {  
        //Statements inside the body of nested "if"  
    }  
    else {  
        //Statements inside the body of nested "else"  
    }  
}  
else {  
    //Statements inside the body of "else"  
}
```

NESTED IF-ELSE EXECUTION SEQUENCE

- If the condition is true, **statements inside the body of if are executed**

- Then this **condition2** will be evaluated.

- If **condition2** is true, **statements inside the body of if are executed**

- If **condition2** is false, **statements inside the body of else are executed**

```
if(condition) {  
    //Nested if else inside the body of "if"  
    if(condition2) {  
        //Statements inside the body of nested "if"  
    }  
    else {  
        //Statements inside the body of nested "else"  
    }  
}  
else {  
    //Statements inside the body of "else"  
}
```

```
if (expression1)
{
    if(expression2)
        statement1;
    else
        statement2;
}
else
{
    if(expression3)
        statement3;
}
```

```
#include<stdio.h>
```

```
void main()
```

```
{    int x;
```

```
    scanf("%d", &x)
```

```
    if(x > 5)
```

```
    {
```

```
        if(x < 10)
```

```
            printf("%d is within 5 and 10", x);
```

```
    }
```

```
}
```

```
#include<stdio.h>
```

```
void main()
```

```
{   int x;
```

```
    scanf(" %d", &x)
```

```
    if(x > 5)
```

```
    {   if(x < 10)
```

```
        printf("%d is within 5 and 10", x);
```

```
    }
```

```
    else
```

```
        printf("%d is not within 5 and 10", x);
```

```
}
```

```
#include<stdio.h>
```

```
void main()
```

```
{   int x;
```

```
    scanf(" %d", &x)
```

```
    if(x > 5)
```

```
        if(x < 10)
```

```
            printf("%d is within 5 and 10", x);
```

```
        else
```

```
            printf("%d is greater than 5 ", x);
```

```
}
```

```
#include<stdio.h>
```

```
void main()
```

```
{   int x;
```

```
    scanf(" %d", &x)
```

```
    if(x > 5 && x < 10)
```

```
    {
```

```
        printf("%d is within 5 and 10", x);
```

```
    }
```

```
    else
```

```
        printf("%d is not within 5 and 10", x);
```

```
}
```

NESTED IF-ELSE

```
if(p)
if(q)
printf("p and q are true");
else
printf("To which if statement does this else belong to?");
```

Ans –

An **else** always associates with the **nearest if** in the **same block** that **does not already have an else** associated with it.

NESTED IF-ELSE Example 1

```
int main(){
    int a;
    printf("enter the number:");
    scanf("%d",&a);
    if(a>0){
        printf("%d is positive number",a);
    }
    else{
        if(a<0){
            printf("%d is negative number",a);}
        else{
            printf("%d is equal to zero",a);}
        }
    }
```

NESTED IF-ELSE Example 2

```
int main(){  
    int a,b,c;  
    printf("enter three numbers:");  
    scanf("%d%d%d",&a,&b,&c);
```

```
        if(a>b){  
            if(a>c){  
                printf("%d is greater",a);}  
            else{  
                printf("%d is greater",c);}  
            }  
        }  
        else{  
            if(b>c){  
                printf("%d is greater", b);}  
            else{  
                printf("%d is greater",c);}  
            }  
        }  
    }
```



Question: UIU Course Picker

Solution: [check out UIU_course_picker.c - provided in drive](#)

EXERCISE 8

Problem:

Write a C program to accept a coordinate point in a XY coordinate system and determine in which quadrant the coordinate point lies. Also find out if the point is on the x or y axis. Or if the point is origin/center point itself!

Sample Input

7 9

-1 0

Sample Output

The coordinate point (7,9) lies in the First quadrant.

The coordinate point (-1,0) is on the X axis.

Solution: [check out coordinate solution.c](#)

Exercise 9

Note: Do On your own. No need to submit. Will randomly ask a few students to share screen and show the code in next class. For use of `getchar()` before `scanf()` while taking char input, refer to the video at time mark 01h:50m:49s

Problem:

Write a C program to take a character. Find out whether the person is a **student** or an **employee**. If he is a **student** then take his **earned credit** as input and find out **which level** he belongs to.

If he is an **employee** then take **another character** as input and find out whether he is **Military** or **Civil Faculty**. If **civil faculty** then again **take another character** as input and find out whether he is **permanent** or **adjunct**.

Following chart has been given for better understanding.

EXERCISE

Character	Earned Credit	Output	
S	0-36	Level 1	
	36-72	Level 2	
	72-108	Level 3	
	>108	Level 4	
Character	Faculty	Output	Output
E	M	Military Faculty	
	C	Civil Faculty	
		Input	
		P	Permanent
		A	Adjunct

Solution: [check out nested conditional logic practice problem.c](#)



This is up to Midterm

*Thank
you!*